AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MS102

Final Examination
Spring Semester 2024/2025
Date: 03/June/2025

Number of Exam Pages: 5 (including this cover sheet)	Time Allowed: 2hrs

Instructions:

- Total Marks: **50**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

AOU EXAM

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PART I: Answer the entire Multiple Choice Questions (MCQ). [20 Marks]

1. What is the gravitational force acting on a 59-kg person due to another 59-kg person standing 2.0 m away? We can model each person as a small sphere. ($G = 6.67 \times 10\text{-}11\ N \cdot m^2/kg^2$)

A) $5.8 \times 10\text{-}8$ N B) 8.5×103 N C) $1.2 \times 10\text{-}7$ N D) $9.8 \times 10\text{-}10$ N

2. A 1200 kg car is traveling at 3.0 m/s and strikes a wall. The car bounces off the wall with a velocity of 4.0 m/s in the opposite direction. If the car is in contact with the wall for 0.10 seconds, then the magnitude of the average force of the wall on the car

a) 46,000 N b) 84,000 N c) 63,000 N d) 75,000 N

3. The units of momentum are

a) ML/T b) M/T c) L/T^2 d) ML/T^2

4. A very elastic rubber ball is dropped from a certain height and hits the floor with a downward speed v. Since it is so elastic, the ball bounces back with the same speed v going upward. Which of the following statements about the bounce are correct? (There could be more than one correct choice.)

A) The ball had the same momentum just before and just after the bounce.
B) The magnitude of the ball's momentum was the same just before and just after the bounce.
C) The ball's momentum was conserved during the bounce.
D) None of the above statements are correct.

5. Three cars, car X, car Y, and car Z, begin accelerating from rest at the same time. Car X is more massive than car Y, which is more massive than car Z. The net accelerating force exerted on each car is identical. After 10 seconds, which car has the most amount of momentum?

A) They all have the same amount of momentum. B) Car X
C) Car Y D) Car Z

6. In an inelastic collision involving an isolated system, the final total momentum is

A) exactly the same as the initial momentum. B) more than the initial momentum.
C) less than the initial momentum D) depends on collision type

7. In a game of pool, the white cue ball hits the #5 ball and stops, while the #5 ball moves away with the same velocity as the cue ball had originally. Both balls have the same mass. This type of collision is

A) Somewhat inelastic. B) Elastic. C) Completely inelastic. D) Semi Elastic

8. A 0.14-kg baseball is dropped from rest from a height of 2.0 m above the ground. What is the magnitude of its momentum just before it hits the ground if we neglect air resistance?

A) 0.28 kg · m/s B) 0.88 kg · m/s C) 0.44 kg · m/s D) 0.62 kg · m/s

9. A 0.10-kg ball, traveling horizontally at 25 m/s, strikes a wall and rebounds at 19 m/s. What is the magnitude of the change in the momentum of the ball during the rebound?

A) 1.2 kg · m/s B) 1.8 kg · m/s C) 4.4 kg · m/s D) 5.4 kg · m/s

10. Which type of wave does not require a medium to propagate?

a) Longitudinal wave b) Transverse wave
c) Mechanical wave d) Electromagnetic wave

11. The rate of change of displacement with respect to time is called:

a) Velocity b) Speed c) Acceleration d) Distance

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12. An object is moving with a constant acceleration. Which of the following statements is true?

a) Its velocity is constant
b) Its acceleration is zero
c) Its velocity is changing at a constant rate
d) Its displacement is zero

13. Sound waves are an example of which type of wave?

a) Transverse wave
b) Electromagnetic wave
c) Mechanical wave
d) Longitudinal wave

14. The intensity of a wave is related to its:

a) Square of Amplitude b) Frequency c) Wavelength d) Speed

15. An object that hangs from the ceiling of a stationary elevator by an ideal spring oscillates with a period T. If the elevator accelerates upward with acceleration 2g, what will be the period of oscillation of the object?

A) 4T B) 2T C) T D) T/2

16. Impulse is defined as the:

a) Change in momentum
b) Force applied over a period of time
c) Product of mass and acceleration
d) Rate of change of velocity

17. The SI unit of force is:

a) Newton b) Kilogram c) Meter d) Second

18. Which of the following is an example of a contact force?

a) Gravitational force
b) Magnetic force
c) Frictional force
d) Electrostatic force

19. The speed of a wave in a string is directly proportional to the:

a) Density of the string
b) Tension in the string
c) Length of the string
d) Wave frequency

20. The distance between two consecutive identical points on a wave is known as its:

a) Amplitude b) Frequency c) Wavelength d) Period

21. If the speed of the ball is doubled without affecting its mass, then the momentum will be

a) 2 times larger. B) 2 times larger. C) 3 times larger. d) Not change.

22. The force required to accelerate an object of mass 4 kg at a rate of 4 m/s^2 is:

a) 16 N b) 1 N c) 8 N d) 0 N

23. The oscillations of particles in a longitudinal wave occur:

a) Perpendicular to the direction of wave propagation
b) In the same direction as the wave motion
c) At right angles to the wave amplitude
d) In random directions

24. A wave passes through a medium at a frequency of 20 Hz. What is the period of the wave?

a) 20 seconds b) 0.05 seconds c) 0.2 seconds d) 2 seconds

25. What is the distance between adjacent nodes in a standing wave?

a) wavelength/4 b) wavelength/3 c) wavelength/2 d) 2 wavelengths

26. When two positively charged objects are brought close together, they will:

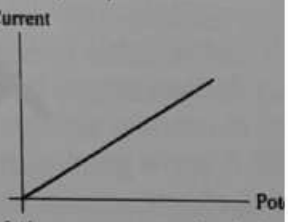
a) Attract each other
b) Repel each other
c) Cancel each other's charge
d) Have no effect on each other

27. Coulomb's law describes the relationship between electric force, charges, and:

a) Time b) Distance c) Speed d) Energy

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28. If the distance between two point charges is doubled, the force between them:
a) Quadruple b) Half c) Double d) Remain unchanged
29. The opposition offered by a material to the flow of electric current is called:
a) Conductivity b) Voltage c) Resistance d) Current
30. Kirchhoff's first law is related to:
a) Conservation of charge b) Conservation of energy
c) Conservation of momentum d) Conservation of voltage
31. What property of an object determines whether it gains or loses electrons?
a) Mass b) Volume c) Electric charge d) Electric field strength
32. The resistance of a wire depends on its:
a) Length only b) Cross-sectional area only
c) Material and length d) Material and cross-sectional area
33. Four 12 Ω resistors are connected together. What is the least resistance that can be attained with these resistors by connecting them in various ways?
a) 12 Ω b) 6.0 Ω c) 3.0 Ω d) 2.0 Ω
34. Two tiny beads are 25 cm apart with no other charges or fields present. Bead A carries 10 μC of charge and bead B carries 1 μC. Which one of the following statements is true about the magnitudes of the electric forces on these beads?
A) The force on A is 10 times the force on B.
B) The force on B is 10 times the force on A.
C) The force on A is exactly equal to the force on B.
D) The force on A is 100 times the force on B.
35. Two identical small charged spheres are a certain distance apart, and each one initially experiences an electrostatic force of magnitude F due to the other. With time, charge gradually diminishes on both spheres by leaking off. When each of the spheres has lost half its initial charge, what will be the magnitude of the electrostatic force on each one?
A) 1/16 F B) 1/8 F C) 1/4 F D) 1/2 F
36. When a current flows through a metal wire, the moving charges are
A) only protons. B) only electrons.
C) both protons and electrons. D) positive metal ions.
37. If a quantity you calculated has units of Ω · m what is that quantity?
A) potential B) resistivity C) resistance D) capacitance
38. For the graph shown in the figure, what physical quantity does the slope of the graph represent for ohmic material?
A) 1/(resistance) B) resistivity
C) 1/(resistivity) D) resistance



39. A wire of resistivity ρ must be replaced in a circuit by a wire of the same material but four times as long. If, however, the total resistance is to remain as before, the diameter of the new wire must
A) be the same as the original diameter. B) be one-half the original diameter.
C) be one-fourth the original diameter. D) be two times the original diameter.

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40. A rod is 4.0 m long and has a square cross-section that is 1.5 cm on each side. An ohmmeter measures 0.040 Ω across its ends. What is the resistivity of the material from which this rod is made?

A) 0.023 Ω·m B) 0.015 Ω·m C) 1.5×10^{-4} Ω·m D) 2.3×10^{-6} Ω·m

Part II: Answer the following questions. [40 Marks]

Question 1: (20 Marks)

a) A string is stretched with a tension of 120 N. The string has a mass of 10.0 grams and has a length of 2.50 meters. Calculate the velocity of wave propagation along the string.(10 Marks)

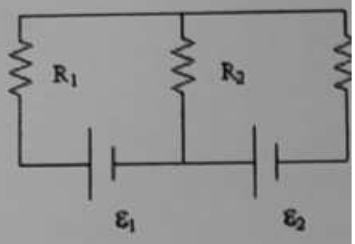
b) Four point charges are located at the corners of a square, 1.00 m by 1.00 m. On each of two diagonally opposite corners are 1.00 μC charges. On each of the other two corners are −1.00 μC charges. What is the magnitude of the force on one of the positive charges? (10 Marks)

Question 2: (20 Marks)

In the diagram, if the current through R_1 is 5 A up, and the current through R_3 is 3 A up,

a) What is the current in R_2? (10 marks)

b) If R1=4Ω, R2=2 Ω, R3=4 Ω What is the equiv resistance of circuit? (10 Marks)



End of questions

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